

Entwurf von Human-AI Interfaces

Task 2

Designing Human-AI Interfaces (193.205)

Group 13

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1 Research papers

1.1 Research papers 1

1.1.1 Journal Article

[1] A. Pradhan, K. Mehta, and L. Findlater, “Accessibility Came by Accident: Use of Voice-Controlled Intelligent Personal Assistants by People with Disabilities,” Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems, (2018).

Our goal in choosing this paper To substantiate the core design decision of using a voice-first interface by demonstrating, through empirical research, that conversational voice systems are genuinely empowering for people with physical, sensory, or cognitive limitations — the very barriers our elderly target group commonly faces.

The main objective of this paper To evaluate the accessibility of off-the-shelf voice-controlled intelligent personal assistants (IPAs), such as the Amazon Echo, through exploratory studies of user reviews and interviews to understand how people with various disabilities use and benefit from conversational interfaces.

Relation to our project Our project targets elderly individuals who may lack technical knowledge or possess physical limitations, utilizing a domestic, voice-controlled interface to remove these barriers. This paper confirms that voice-controlled IPAs greatly expand accessibility, allowing users to independently complete daily tasks, alleviate caregiver burden, and receive memory support. It validates your design choice by showing that conversational interfaces offer “tremendous potential for inclusive, accessible interaction” without requiring complicated screens

1.1.2 Journal Article

[2] K. K. Fitzpatrick, A. Darcy, and M. Vierhile, “Delivering Cognitive Behavior Therapy to Young Adults With Symptoms of Depression and Anxiety Using a Fully Automated Conversational Agent (Woebot): A Randomized Controlled Trial,” JMIR Mental Health, Vol. 4, No. 2, (2017).

Our goal in choosing this paper To prove that a fully automated, AI-driven conversational agent can successfully build an engaging relationship with users and yield measurable improvements in emotional well-being.

The main objective of this paper To determine the feasibility, acceptability, and preliminary efficacy of using a text-based conversational agent (Woebot) to deliver cognitive behavioral therapy (CBT) and emotional support through brief, daily interactions.

Relation to our project Our solution relies on natural language processing to hold open-ended conversations responding to the user’s emotional needs. This research establishes that text-based conversational agents (Woebot) successfully deliver cognitive behavioral therapy, increase emotional awareness, and significantly reduce symptoms of depression in a cohort of young adults aged 18 to 28. The project adapts this scientifically validated conversational intervention framework from a young adult demographic to a geriatric demographic. It validates the premise that an AI companion fosters accountability and is perceived as an empathetic presence rather than standard medical equipment.

1.1.3 Conference Proceedings

[3] J. Fogarty, S. E. Hudson, and J. Lai, “Examining the Robustness of Sensor-Based Statistical Models of Human Interruptibility,” Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, (2004).

Our goal in choosing this paper To provide theoretical support for your system’s passive

continuous monitoring, proving that background sensors can effectively and non-intrusively track user activity to inform proactive interventions.

The main objective of this paper To examine the effectiveness and robustness of sensor-based statistical models—specifically using microphones and activity trackers—in predicting human interruptibility and availability across different environments without relying on active user reporting.

Relation to our project A key feature of our AI Emotional Companion is its ability to proactively initiate check-ins rather than waiting passively for user commands — an important distinction for elderly users who may not always think to reach out when they need support. This paper validates the technical feasibility of that feature: it shows that a system can analyze ambient acoustic signals to infer when a person is unoccupied and receptive to interaction, without requiring them to press a button or speak first. We apply this principle to our device, where equivalent acoustic sensing informs the timing of proactive companionship interactions. This passive sensing capability furthermore operates independently of the emotion detection layer, meaning the two systems — one determining when to engage, the other determining how to respond emotionally — can function in parallel within a coherent architecture.

1.1.4 Reflection

Tools used: Gemini Pro Claude (Anthropic)

To find relevant sources, I prompted Gemini Pro to search through abstracts and suggest keywords aligned with our project's research goals. After identifying suitable papers, I used Gemini to gain an initial understanding of their methodology and findings, then manually cross-checked its summaries against the original papers to ensure accuracy. I then used Claude to improve the quality and academic accuracy of my reference entries — rewriting sections for clearer structure, stronger argumentation, and more precise language. Claude also helped identify an overclaim in Source 3's "Relation to our project," which I corrected and verified manually.

1.2 Research papers 2

1.2.1 Journal Article

Holt-Lunstad J, Smith TB, Baker M, Harris T, Stephenson D. **Loneliness and social isolation as risk factors for mortality: a meta-analytic review.** *Perspect Psychol Sci.* 2015 Mar;10(2):227-37.

Our goal in choosing this paper is to highlight that loneliness carries not only an emotional burden, but a fatal risk for peoples' health. Based on numerous studies, the research source proves a measurable link between social isolation and increased mortality rates.

The main objective of this paper is to examine how social isolation, loneliness, and living alone increase the risk for early mortality using a meta-analysis. For this purpose, the article analyzes cumulative data from 70 independent studies with nearly 3.5 million participants, followed for an average of 7 years.

The paper distinguishes between loneliness - the subjective feeling of isolation, disconnectedness and not belonging; social isolation - the objective lack of contact and communication; and living alone as three separate factors. Results show that the level of risk is similar across all three factors, no matter whether loneliness is defined as objective or subjective.

The studies' findings remain consistent across different groups in terms of age, gender, and physical health. The effects of social isolation are compared to other major health risk factors like obesity, sedentary lifestyle, smoking, and air pollution.

In conclusion, the authors highlight social connection as a fundamental factor for human survival and call for stronger policy attention to prevent loneliness from reaching epidemic proportions till the end of this decade.

Relation to our project: This source defines the core problem our project addresses and highlights its gravity. It supports the need for our AI Emotional Companion by showing the direct connection between loneliness and decaying health, and how a lack of socialization and meaningful interactions negatively affect ones' well-being.

1.2.2 Journal Article

Dino MJ, Leinbach C, Dino G, Thiamwong L, Villafuerte CM, Shattell M, Pimentel J, Zamora MA, Bautista A, Vitug JP, Chepkorir J, Marave N. **Smart Speakers for Health and Well-Being of Older Adults: A Mixed-Methods Review.** Healthcare (Basel). 2025 Oct 31;13(21):2772.

Our goal in choosing this paper this paper is to evaluate the effectiveness of smart speaker systems for older people and to analyze their advantages and limitations. Through this source, we aim to extract proven effective aspects that can be applied in our solution, as well as to identify the barriers this technology faces and how we can overcome them in order to develop a more accessible solution for the elderly.

The main objective of this paper is to analyze the impact of voice-activated intelligent assistants on the health of older adults. The study applies a mixed-methods approach, combining statistics data and user experience reports, and focuses on people over the age of 65 who often face social challenges, including social isolation and reduced mobility.

Findings show improvements in daily routine, emotional well-being, social connections, and medication adherence, but also point out challenges like privacy concerns, technical difficulties, cost, and limited digital literacy. While the AI companion is shown to ease loneliness, limitations in conversational ability and personalization as well as the mismatch between the technological competency of older adults and the technological requirements of smart speakers restrict its effectiveness.

Overall, the authors highlight the strong potential of smart assistants to enhance the physical, mental, and social health of older adults, but emphasize the need for better accessibility and user support.

Relation to our project: This paper is important for our project, as it confirms that the use of smart devices based on voice input can have a real positive effect on the well-being of older people. On the one hand, the study shows which functionalities are most effective and can be integrated into our solution - small daily interactions with the companion, reminders about taking medication, daily activities such as playing music and contacting a family member. On the other hand, it introduces us to the main problems the technology faces, which gives us a clear direction on what we need to improve to make our solution more user friendly.

1.2.3 Conference Proceedings

Sharmila G. **Speech Emotion Recognition: Features and Classification Models Using CNN.** 4th International Conference on Inventive Research in Computing Applications (ICIRCA). Coimbatore, India. 2022;1087-1091.

Our goal in choosing this paper is to explore the possibilities of integrating emotion recognition from speech into our project. The main challenge before us is to find a technology that is as adaptive as possible to the emotional needs and state of the elderly users. This source introduces us to the so-called speech emotion systems and how these models handle audio and classify it to a specific emotional state, as well as the main limitations of the technology.

The research is based on the Convolutional Neural Network (CNN) for speech emotion recognition. The model is compared with three other speech emotion algorithms. Results show that CNN reaches the highest level of sensitivity.

The model achieves this by extracting the features from the audio signal in order to associate it with an emotion. It is important to note, however, that the system comes with serious challenges such as the recording setting, which can lead to quite different results when it is not controlled as a part of an experiment, but part of the user's daily reality. Speaker variance, and acoustical likeness of one emotion to another are other main concerns of the author.

Relation to our project: The research is important for our project, as this highly intelligent technology layer is what would differentiate our solution from ordinary smart assistants or similar systems. Recognizing the user's emotions would allow our system to not only execute simple commands and act as a daily assistant / reminder, but to offer real emotional support and enrich the lives of elderly people isolated from society.

1.2.4 Reflection

Tools used: Chat GPT 5.3

Reflection: I used the AI tool for an initial summary of the research papers. This helped me reach the most relevant ones in the context of our project and research goals. It was not very useful for the preparation of the text and more precisely these parts in which I present the main contents of the papers, as it often relied on the abstract of the paper instead of delving deeper into its entire content. I wrote the text myself, looking for key points in the paper with the help of the AI's summaries. I also used AI to understand the topic of the last paper better as it is more technical oriented.

1.3 Research papers 3

1.3.1 Journal Article 7

Y. Deng, L. Liao, W. Lei, G. H. Yang, W. Lam, and T. Chua, "Proactive Conversational AI: A Comprehensive Survey of Advancements and Opportunities," *ACM Transactions on Information Systems*, Vol. 43, No. 3, (2025).

Our goal in choosing this paper To provide strong theoretical backing for our system's core differentiator: shifting from a standard reactive smart speaker to a proactive AI companion.

The main objective of this paper To offer a comprehensive review of proactive dialogue systems, which empower conversational agents to take the initiative, anticipate user needs, and purposefully guide conversations toward specific functional or social goals, rather than merely passively following user commands.

Relation to our project Unlike existing general-purpose smart speakers, our companion is proactive and initiates check-ins based on inferred emotional states. This paper directly supports this design through its analysis of "**emotional support dialogues**," where a proactive agent does not just echo negative feelings, but takes the initiative to explore the user's distress and strategically suggests supportive interventions—such as playing music or recommending social activities—to actively lower their emotional distress

1.3.2 Journal Article 8

B. Yu, J. R. Williamson, J. C. Mundt, and T. F. Quatieri, "Speech-Based Automated Cognitive Impairment Detection From Remotely-Collected Cognitive Test Audio," *IEEE Access*, Vol. 6, (2018).

Our goal in choosing this paper To offer robust clinical and technical validation for the “audio analysis layer” of our device, establishing that passive, continuous vocal monitoring is a scientifically proven assessment method.

The main objective of this paper To investigate the effectiveness of extracting non-linguistic speech features—specifically phoneme speaking rates, pitch variability, and articulatory coordination—from remotely collected audio to automatically detect cognitive impairment and psychomotor slowing in the elderly.

Relation to our project Our system continuously analyzes vocal cues such as tone, speaking pace, and unusual silence to detect early signs of cognitive decline without asking the user to report how they feel. This paper confirms the validity of this approach by proving that machine learning models accurately detect cognitive decline and psychomotor slowing via phoneme duration and pause analysis. The validation for the psychological (depression) monitoring is secured by the Amir-Behghadami et al. (2026) source, isolating the application of this specific paper strictly to the detection of neurological impairment.

1.3.3 Journal Article 9

D. H. Gustafson Sr, M.-L. Mares, D. C. Johnston, G. Landucci, K. Pe-Romashko, O. J. Vjorn, Y. Hu, D. H. Gustafson Jr, A. Maus, J. E. Mahoney, and B. Mutlu, “Using Smart Displays to Implement an eHealth System for Older Adults With Multiple Chronic Conditions: Protocol for a Randomized Controlled Trial,” *JMIR Research Protocols*, Vol. 11, No. 5, (2022).

Our goal in choosing this paper To scientifically justify the physical form factor, hardware choice, and simplified user interface of our proposed solution.

The main objective of this paper To detail a randomized controlled trial designed to test whether delivering an evidence-based eHealth intervention to older adults via a voice-controlled smart display is more effective at improving quality of life and health outcomes than delivering the same intervention via a standard touch-screen laptop.

Relation to our project Our project takes the physical form of a small smart display with a built-in speaker, a large-text touchscreen, and a voice-first interface. This paper directly validates this specific hardware choice, noting that **voice-controlled smart displays are uniquely suited for older adults who often face physical limitations—like hand tremors or vision problems—that make traditional computers and smartphones incredibly difficult to use**. It supports our approach of combining a domestic design with a very simple, uncluttered visual and voice interface to ensure the user never feels overwhelmed

1.3.4 Reflection

Tool Used: Gemini Pro & NotebookLM

I have used Gemini Pro to generate ideas on how and what to search for to find as accurate sources as possible. Then uploaded them to NotebookLM to understand the papers, specifically interact and generate texts with citations. Rest of the work was to cross-check the generated texts with the relevant citations.

2 Context

Socio-Technical Environment:

Areas of Life: Primarily the **private domestic sphere** (the home) and **assisted living facilities**. It touches upon healthcare (medication/appointments), social connectivity (family relations), and entertainment.

Physical Environment: A living space equipped with internet connectivity and voice-activated hardware (smart speakers, tablets, or dedicated robotic interfaces). It must be an environment where the user feels safe to speak aloud.

Usage Process: The interaction is **passive-to-active**. The AI remains in a “listening” or “observing” state (monitoring patterns) and transitions to an active state through voice triggers or proactive check-ins.

2.1 Primary Challenges

- **The Digital Divide:** Many elderly users are intimidated by traditional screens, logins, and complex menus, leading to a “tech-phobia” that prevents them from using helpful tools.
- **The Silent House:** Long periods of silence lead to cognitive decline. The lack of linguistic stimulation can accelerate memory loss and depression.
- **Fragmented Care:** Family members and doctors often only see a “snapshot” of the person during a weekly visit, missing the daily fluctuations in mood or health.
- **Cognitive Load:** Managing multiple medications and a calendar of appointments becomes a source of high stress for those with early-stage cognitive impairment.

2.2 Importance of the Solution Approach

- **Natural Language as an Equalizer:** By using voice, the barrier of the digital divide is removed. If a user can talk, they can use the system.
- **Proactive vs. Reactive:** Unlike a traditional emergency button that requires the user to act during a crisis, our AI monitors **speech patterns** (tone, speed, pauses) to detect depression or health issues *before* they become emergencies.
- **Human-Centric Design:** It treats the user as a person with emotional needs, not just a patient with medical requirements. Suggesting a “favorite song” is as important as a “medication reminder” for maintaining the spirit.

2.3 Why the Problem is Critical in This Context

- **Physical Health Impact:** Studies show that chronic loneliness can be as damaging to health as smoking 15 cigarettes a day. It weakens the immune system and increases the risk of heart disease.
- **The Feedback Loop of Isolation:** A lonely person is less likely to exercise or eat well, which leads to physical decline, which leads to further isolation.
- **Economic Strain:** Unmanaged loneliness leads to more frequent hospitalizations and earlier transitions to expensive professional nursing care. By supporting aging in place, our solution provides a sustainable economic alternative for families and the healthcare system.
- **Safety Risks:** A missed medication or a fall that goes undetected for 24 hours is a life-threatening event. In an isolated environment, the “cost of silence” can be fatal.

3 Persona

3.1 Persona 1: The Primary User (Archetypal)



Name: Martha Higgins

Age: 82 years old

Location: Small apartment in a retirement community.

Character: Martha is a retired schoolteacher who loves crossword puzzles and opera. She is fiercely independent but has recently started forgetting her blood pressure medication.

Personality: Martha feels a deep sense of quietness in her home. Touchscreens frustrate her because of her arthritis. She values politeness and human-like interaction.

Project Connection: She uses the AI for cognitive stimulation. When the AI detects her speech slowing down or becoming repetitive (signs of fatigue or low mood), it prompts her to listen to her favorite opera songs, which significantly boosts her daily motivation.

3.2 Persona 2: The Secondary Persona (Stakeholder)



Name: James Higgins

Age: 49 years old

Role: Martha's son and a busy Project Manager.

Character: James is tech-savvy and constantly on the move. He lives three hours away from his mother.

Personality: He suffers from passive worrying anxiety. He worries about his mother falling or missing pills but doesn't want to piss her off with daily calls that make her feel monitored.

Project Connection: James is the "Administrator" stakeholder. He relies on the AI to be his eyes and ears. He values the emergency notification feature and the weekly Emotional Health Report the AI generates based on his mother's interactions.

3.3 Persona 3: The Non-Persona



Name: Chloe Vance

Age: 22 years old

Character: University student and freelance graphic designer.

Personality: Highly tech-literate, prefers text over voice, and seeks efficiency over companionship from her devices.

Why she is a Non-Persona: Chloe uses AI for productivity (scheduling, coding, automation). She finds proactive conversational AI annoying and intrusive. Our solution's slow-paced, empathetic, and voice-only nature would be inefficient for her lifestyle.

3.3.1 Reflection

Tools Used: Gemini 3 Flash & Manual Content Editing.

Reflection: The AI tool was good for generating the foundational information of the characters like names, ages, photos and general life situations, which saved some time. However, the AI tended to make the Secondary Persona (James) too focused on himself. I had to adapt his profile to show how he interacts with the data our project provides. For the Non-Persona, the AI initially suggested a child, but I changed this to a young adult because it more clearly defines the technical boundaries of your voice-first, slow-interaction model, and we can be sure that the person is capable of using AI tools and knows how to use them.

4 Solution

4.1 Technical Approach

The proposed solution is an AI-powered emotional companion designed specifically for elderly people who live alone or have limited social contact. The system takes the form of a small desktop device, similar to a smart display with a built-in speaker, that sits in a central spot in the user's home like the living room or kitchen. It is always on and ready to use, with no technical knowledge required.

The main way to interact with it is through voice: the user simply speaks to the device and it responds in a calm, friendly tone. There is also a touchscreen for users who prefer tapping over talking, and a clearly visible emergency button for urgent situations.

4.2 How AI Contributes to the Problem

AI is what makes this solution genuinely useful in everyday life. Unlike a basic reminder app or a general-purpose smart speaker such as Amazon Echo or Google Nest, this system is built entirely around the emotional and practical needs of elderly people. Where existing devices require the user to initiate every interaction, this companion is proactive: it checks in, notices changes, and responds to the user's emotional state rather than just their commands.

The system uses natural language processing to hold real, open-ended conversations. Beyond that, it continuously analyzes vocal cues such as tone, speaking pace, and response latency to detect early signs of emotional change, for example unusual silence, a flatter tone than normal, or shorter answers than usual. When a pattern like this is detected, the system does not simply log it: it responds by gently suggesting a social activity, playing familiar music, or recommending the user contact a family member or doctor.

This kind of passive, continuous emotional monitoring is what sets the system apart from existing solutions. It does not require the user to report how they feel: it infers it, and acts on it.

Regarding privacy, all voice processing is handled on-device wherever possible, so conversations are not stored in external servers. The system only transmits data when strictly necessary, for example when placing a call or alerting an emergency contact. Users and their families are informed of exactly what data is collected and can adjust these settings at any time.

4.3 Shape and Appearance of the Artifact

As shown in Figure 1, the device resembles a small smart display mounted on a short stand, placed on a table or shelf. The screen is large relative to the frame, with a speaker integrated on one side. The physical design is intentionally neutral and domestic, so it blends into a home environment without feeling like medical equipment.

The interface uses large text, high contrast, and very few elements per screen so it is easy to read and navigate. There are four key screens: the home screen, the hourly check-in, the medication reminder, and the navigation menu. All of them follow the same principle of simplicity, making sure the user never feels overwhelmed.

4.4 Planned Forms of Interaction

The system supports three main types of interaction. The first is voice conversation, where the user can talk to the device about anything, from how they are feeling to what they did that day. The second is reminders, where the device proactively notifies the user about medication, appointments, or daily activities at set times. The third is emergency support, where the user

can press a clearly visible button or ask the device to call a doctor or a family member at any moment.

The hourly check-in, visible in screen (b) of Figure 1, asks a simple question like “How are you feeling right now?” and lets the user respond either by voice or by tapping one of four emotion options. This low-effort interaction is the main way the system tracks emotional state over time without placing any burden on the user.

4.5 Required Technical Components

At a high level, the system relies on a few key components: a microphone and speaker for voice interaction, a touchscreen for the visual interface, a speech recognition module to convert voice into text, and a language model to generate natural responses. An additional audio analysis layer monitors vocal features for emotional cues. The device also connects to the internet to enable calls to family members or emergency contacts when needed.

4.6 Why This Solution Makes Sense

Elderly people who experience loneliness rarely need complex technology. What they need is something that listens, reminds, and reaches out when necessary. This solution is designed around those three needs. It is simple enough that no instruction manual is needed, but smart enough to notice when something might be wrong. By combining emotional support with practical daily assistance in a single, approachable device, it fits naturally into the life of an elderly person without requiring any change in their routine.

4.7 Reflection

Tools Used: Claude

Reflection: I used this tool to gather information and research relevant areas of this project. I wrote all the content myself and used Claude to review and validate that the concepts were accurate and aligned with my research. Where this tool was most helpful in creating the prototype diagram, which was done almost entirely by Claude, I then reviewed it and adjusted some details to better match our vision.

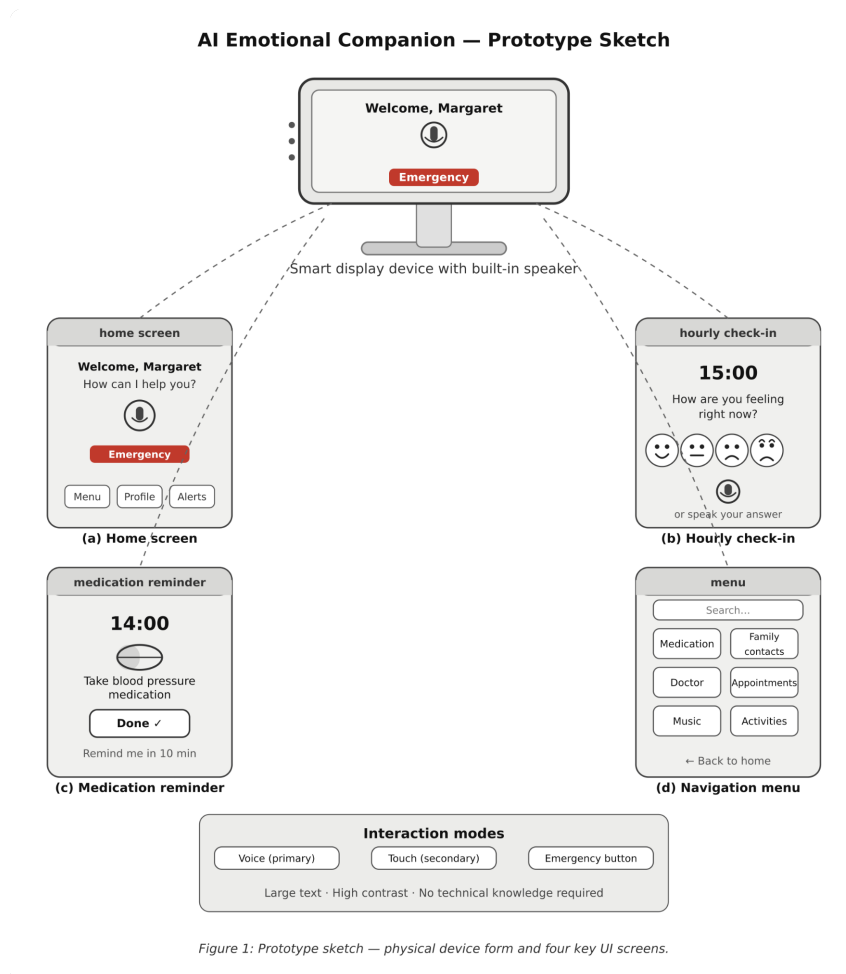


Figure 1: Prototype sketch — physical device form and four key UI screens.